Achintya Rai

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Education

University of California, San Diego — Sept. 2024 – June 2028
Bachelor of Science in Computer Engineering — ***GPA: 3.9/4.0***

- Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow

Work Experience

VALT Health — Remote
Founding Software Engineer — *Aug. 2025 – Feb. 2026*

- Engineered a scalable Flask backend to process **100k+** datapoints from fitness devices (Fitbit, Apple Watch), designing data structures and database schemas to synchronize formats across **20+** distinct sources.
- Deployed concurrent batch processing pipelines on distributed systems in GCP, achieving a **35%** throughput increase and **60%** reduction in redundant database write operations.
- Automated local environment setup via Bash scripts, reducing debugging and testing time by up to **80%** and improving end-to-end infrastructure reliability for the engineering team.

Decklar (formerly Roambee) — Sunnyvale, CA
Software Engineering Intern — *June 2024 – Aug. 2024*

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, implementing advanced data structures and algorithms with Python and NumPy to achieve a **15%** capacity boost for **1M+** weekly events.
- Engineered high-performance analytics modules with systematic debugging of data integrity issues, delivering **20%** faster reporting speeds across large-scale systems processing **1M+** weekly events.

Research

Graph Theory & Neural Architecture — San Diego, CA
Undergraduate Research Fellow (NSF REU) — *March 2026 – Present*

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction algorithms to compress large-scale systems data from **500GB** checkpoints to **13GB** localized manifolds.
- Optimized analysis scripts for artificial intelligence model backbones via Louvain Clustering, achieving a **30x runtime reduction** (30 mins to <1 min) on resource-constrained **8GB RAM** systems.

Projects

NanoGPT | *Python, PyTorch, C++, Transformers, LLM Optimization* | — Mar. 2026

- Implemented a GPT-style transformer in Python and PyTorch, engineering algorithms for multi-headed attention, causal masking, and backpropagation; built scalable dataloaders to stream tokenized datasets, maintaining **100%** hardware utilization across training runs.
- Optimized model memory layout and compiled critical sub-modules in **C++** to produce high-performance AI-integrated software, reducing per-token latency by **25%** on resource-constrained systems.

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* — Aug. 2025 – Dec. 2025

- Architected end-to-end infrastructure on AWS for real-time coach–student session management, implementing **20+ Django REST API endpoints** for secure authentication, concurrency-safe session handling, and transaction logic.
- Deployed WebSockets-based synchronization for real-time lesson tracking across concurrent users, establishing a CI/CD pipeline with automated testing to ensure **99.9%** deployment reliability.

Technical Skills

Languages: Python, C/C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS
Development: React, Django, Node.js, FastAPI, Asyncio, WebSockets, Flask, JUnit, PyTest
AI & Cloud: PyTorch, AWS, GCP, Gemini/OpenAI API, Docker, GitHub Actions, Git, Linux